

AI Act Mini Report

Classification

Based on the provided specifications of the EU AI Act, this algorithm is classified as a *high-risk algorithm*. The algorithm falls directly under Annex III, Section 3(b), which states: "*AI systems intended to be used to evaluate learning outcomes, including when those outcomes are used to steer the learning process of natural persons in educational and vocational training institutions at all levels*". Predicting whether a student might drop out fits in the rationale of "steering the learning process" and therefore is classified as a high-risk algorithm, despite other qualifications. Article 6 allows for exceptions if an AI system performs narrow procedural tasks, improves on human workflows, or acts as a support tool without replacing the human in the loop. However, these exceptions are nullified by a condition in the same Article: "*Notwithstanding the first subparagraph, an AI system referred to in Annex III shall always be considered to be high-risk where the AI system performs profiling of natural persons*".

While the algorithm is classified as High-Risk, care must be taken to ensure it does not shift to Prohibited AI classifications (Article 5). A relevant statement which must be taken into account is that the employed system may not exploit any of the vulnerabilities of a natural person or specific group of persons due to their age, disability or a specific social or economic situation, with the effect of harming this person or group of people. Therefore, it is critical to take into account the variables and proxies for age, disability and economic situation (debtor/scholarship status) and approach fairness constraints with care and to always keep a human in the loop.

Safeguards

To deploy this High-Risk system legally and ethically, the following safeguards (Chapter III, Section 2 of the AI Act) must be implemented.

Risk Management System

- A continuous, iterative process to identify and mitigate risks throughout the AI's lifecycle must be implemented.

Data Governance

- Training, validation, and testing datasets must be relevant, representative, free of errors, and complete. Bias examination is a requirement.

Technical Documentation

- Detailed documentation about the logic of the algorithm must be created before the system enters the market, allowing authorities to assess compliance.

Record-Keeping / Logging

- Automatic generation of logs when the algorithm is put to work to ensure traceability of the system's functioning.

Transparency & Instructions for Use

- The system must be interpretable by users (administrators/faculty/student councillors) and accompanied by clear instructions.

Human Oversight

- Design the system so natural persons can oversee, intervene, or stop the system; specifically, the system must not enforce any action by itself.

Accuracy, Robustness, & Cybersecurity

- The system must achieve appropriate levels of accuracy and fairness for all demographic groups and resist errors or malicious attacks.